Middle-Rung Law: physics and chemistry formulas written with the ether frequency v_e and the expansion rate H

Companion file to the OSF preregistration. Predictions fixed 2026-11-05. Author: John Lu Qianhe (约翰·路谦和). Normalization relation $4\pi^2r^3c^2/(T^2Ghv)=1$ after Zhang Xiangqian (2026, unpublished). All numbers use CODATA 2018 constants.

Middle-rung law. At every epoch t:
$\hbar \cdot G = c^5 / (4\pi^2 \cdot v_e(t) \cdot H(t))$, equivalently $v_e(t) \cdot H(t) = v_P^2$, with $v_P = \sqrt{(c^5/\hbar G)}/2\pi$ the Planck (human-boundary) frequency.
v_e and H both change with cosmic time; their product is fixed, so ħ and G stay constant.

0. Axioms and definitions (author, 2026-11-05/06)

1. **The universe consists only of ether structures and ether motion.** Different ether structures are what humans call different kinds of matter. Humans are ether structures too.
2. **Basic ether structure:** the regular tetrahedron. Every other structure (all matter) is a knot of it.
3. **Ether space around every structure:** the basic ether circulates around each ether structure in a right-hand motion — like four fingers curling counterclockwise around the thumb, spiralling outward at light speed, while also moving at light speed along the thumb axis. Rotation gives two dimensions, advance gives the third. Every structure, without exception, carries such a surrounding ether space.
4. **Relations between structures:** forces (gravitational, electromagnetic, …) are field-pressure differences where the ether spaces of different structures overlap. Flow contributions add in quadrature ($v^2 = 2\Phi$ superposes linearly); a uniform ether produces no force.
5. **Light speed is the background; only deviations are measurable.** The measurable part around a mass is the deviation flow $\sqrt{(2GM/r)}$ (11.2 km/s at Earth's surface). An outward flow of this speed has fluid acceleration $v \cdot dv/dr = -GM/r^2$, identical to an inward flow, so the outward-spiral reading reproduces the same exterior (Schwarzschild) geometry and all solar-system tests.
6. **Elements:** ether structure n = the element with atomic number n (1 = H … 118 = Og); every substance is “ether structure number + frequency”, with frequency $f = mc^2/h$.

Open points: how ether is created/absorbed at a structure (continuity); whether the thumb axis corresponds to spin/magnetic moment.

1. Present values (H_0 = 61.8–63.5 km s^{-1} Mpc^{-1}; central 62.0)

Quantity	Central (H_0 = 62.0)	Range (61.8–63.5)
Expansion rate H (universe rung)	2.009×10^{-18} s^{-1}	2.003×10^{-18} – 2.058×10^{-18} s^{-1}

Ether frequency $v_e = v_P^2/H$	4.337×10^{102} Hz	$4.235\times10^{102} – 4.351\times10^{102}$ Hz
Planck (human-boundary) frequency v_P	2.952×10^{42} Hz	constant
Invariant product $v_e \cdot H = v_P^2$	8.715×10^{84} Hz²	constant
Rung ratio $N = v_P/H$	1.469×10^{60}	$1.435\times10^{60} – 1.474\times10^{60}$
Universe age ($H_0 t_0 = 0.876$)	13.82 Gyr	13.49–13.86 Gyr
Mid-rung frequency $\sqrt{(H\, v_P)}$	2.435 THz	2.43–2.46 THz

Numerically: $\hbar \cdot G = 7.039\times10^{-45}$ J·m³ kg⁻¹ s⁻² = $c^5 / (4\pi^2 \cdot 4.337\times10^{102} \cdot 2.009\times10^{-18})$.

2. Physics formulas — symbolic form (v_e, H vary with time) and present-value form

Formula	Standard	Ether form (time-dependent $v_e(t)$, H(t))	Present-value form ($v_e \cdot H = 8.71\times10^{84}$ Hz²)	Value
Mass–energy	$E = mc^2$	$E = mc^2$ (unchanged); knot frequency $f = mc^2/h$	$E = mc^2$	8.988×10^{16} J ($m = 1$ kg)
Planck relation	$E = hv$	$E = c^5 v / (2\pi G\, v_e H)$	$E = c^5 v / (2\pi G \cdot 4.34\times10^{102} \cdot 2.01\times10^{-18})$	6.626×10^{-19} J ($v = 10^{15}$ Hz)
de Broglie wavelength	$\lambda = h/p$	$\lambda = c^5 / (2\pi G\, v_e H\, p)$	$\lambda = c^5 / (2\pi G \cdot 8.71\times10^{84} \cdot p)$	7.274×10^{-10} m (electron, 10^6 m/s)
Newtonian gravity	$F = GMm/r^2$	$F = c^5 M m / (4\pi^2 \hbar\, v_e H\, r^2)$	$F = c^5 M m / (4\pi^2 \hbar \cdot 8.71\times10^{84} \cdot r^2)$	9.82 N (1 kg at Earth surface)
Ether inflow = escape speed	$v = \sqrt{(2GM/r)}$	$v = \sqrt{(c^5 M / (2\pi^2 \hbar\, v_e H\, r))}$	$v = \sqrt{(c^5 M / (2\pi^2 \hbar \cdot 8.71\times10^{84} \cdot r))}$	11,190 m/s (Earth surface)
Gravitational redshift	$\Delta f/f = GM/(rc^2)$	$\Delta f/f = f_M f_r / (v_e H)$, $f_M = Mc^2/h$, $f_r = c/2\pi r$	$\Delta f/f = f_M f_r / 8.71\times10^{84}$	6.961×10^{-10} (Earth surface)
Schwarzschild radius	$r_s = 2GM/c^2$	$r_s = c^3 M / (2\pi^2 \hbar\, v_e H)$	$r_s = c^3 M / (2\pi^2 \hbar \cdot 8.71\times10^{84})$	2954 m (Sun)
Planck length	$\ell_P = \sqrt{(\hbar G/c^3)}$	$\ell_P = c / (2\pi \sqrt{(v_e H)})$	$\ell_P = c / (2\pi \sqrt{(8.71\times10^{84})})$	1.616×10^{-35} m
Planck time	$t_P = \sqrt{(\hbar G/c^5)}$	$t_P = 1 / (2\pi \sqrt{(v_e H)})$	$t_P = 1 / (2\pi \sqrt{(8.71\times10^{84})})$	5.391×10^{-44} s
Planck mass	$m_P = \sqrt{(\hbar c/G)}$	$m_P = 2\pi \hbar \sqrt{(v_e H)} / c^2$	$m_P = 2\pi \hbar \sqrt{(8.71\times10^{84})} / c^2$	2.176×10^{-8} kg
Black-hole entropy	$S = k_B c^3 A / (4\hbar G)$	$S = \pi^2 k_B\, v_e H\, A / c^2$	$S = \pi^2 k_B \cdot 8.71\times10^{84} \cdot A / c^2$	1.321×10^{46} J/K ($A = 1$ m²)

Hawking temperature	$T = \hbar c^3 / (8\pi G M k_B)$	$T = \pi \hbar^2 \nu_e H / (2c^2 M k_B)$	$T = \pi \hbar^2 \cdot 8.71\times10^{84} / (2c^2 M k_B)$	6.168×10^{-8} K (solar-mass black hole)
Fine-structure constant	$\alpha = e^2/(4\pi\varepsilon_0\hbar c)$	$\alpha = \pi e^2 G \nu_e H / (\varepsilon_0 c^6)$	$\alpha = \pi e^2 G \cdot 8.71\times10^{84} / (\varepsilon_0 c^6)$	7.297×10^{-3} (dimensionless; e = electron charge)
Bohr radius	$a_0 = 4\pi\varepsilon_0\hbar^2 / (m_e e^2)$	$a_0 = \varepsilon_0 c^{10} / (4\pi^3 G^2 \nu_e^2 H^2 m_e e^2)$	$a_0 = \varepsilon_0 c^{10} / (4\pi^3 G^2 (8.71\times10^{84})^2 m_e e^2)$	5.292×10^{-11} m
Stefan–Boltzmann constant	$\sigma = \pi^2 k_B^4/(60\hbar^3c^2)$	$\sigma = \pi^2 k_B^4 (4\pi^2 G \nu_e H)^3 / (60 c^{17})$	$\sigma = \pi^2 k_B^4 (4\pi^2 G \cdot 8.71\times10^{84})^3 / (60 c^{17})$	5.670×10^{-8} W m^{-2} K^{-4}
Hubble law	$v = H d$	$v = H d$ (H = universe-rung frequency)	$v = 2.01\times10^{-18}$ s^{-1} · d	62,000 m/s (d = 1 Mpc)
Normalization (Zhang 2026) on Hubble sphere	$4\pi^2r^3c^2/(T^2Ghv) = 1$	$r = c/H$, $T = 2\pi/H \Rightarrow v = c^5/(hGH) = 2\pi \nu_e$	$v = 2\pi \cdot 4.34\times10^{102}$	2.725×10^{103} Hz
Temperature → frequency	$\nu = k_BT/h$	$\nu = 2\pi k_B T G \nu_e H / c^5$	$\nu = 2\pi k_B T G \cdot 8.71\times10^{84} / c^5$	6.459×10^{12} Hz (T = 310 K)
Two knots: gravity	$F = Gm_1m_2/r^2$	$F = c h f_1 f_2 / (2\pi \nu_e H r^2)$	$F = c h f_1 f_2 / (2\pi \cdot 8.71\times10^{84} \cdot r^2)$	6.674×10^{-11} N (two 1 kg, 1 m)
Electric / gravity ratio	$F_e/F_g = e^2/(4\pi\varepsilon_0 G m_1 m_2)$	$F_e/F_g = \alpha \nu_e H / (f_1 f_2)$	$F_e/F_g = \alpha \cdot 8.71\times10^{84} / (f_1 f_2)$	1.236×10^{36} (two protons)
Electric = gravity crossover	Stoney mass $\sqrt{\alpha}\, m_P$	$f = \sqrt{(\alpha \nu_e H)}$, $m = h f / c^2$	$f = \sqrt{(\alpha \cdot 8.71\times10^{84})}$	1.859×10^{-9} kg

Each ether form is an algebraic rewriting via the middle-rung law; the value column was checked to agree with the standard form to 10^{-9}. The physical content lies in the interpretation and in the dated predictions (P1–P5), not in the numerical agreement.

3. Chemistry formulas (natural constant e appears only in exponentials and logarithms)

Formula	Form
Ideal gas	$PV = nRT$, $R = N_A k_B$
Boltzmann distribution	$n_2/n_1 = e^{-h\Delta\nu/k_BT}$
Arrhenius	$k = A\, e^{-E_a/RT}$
Nernst	$E = E^0 - (RT/zF) \ln Q$; $2.303\, k_BT/e = 61.5$ mV per decade at 310 K
Gibbs / equilibrium	$\Delta G^0 = -RT \ln K$, $K = e^{-\Delta G^0/RT}$; $\Delta G = -nFE$
Proton-motive force	$\Delta p = \Delta\psi - (2.303RT/F)\, \Delta pH$
Radioactive decay	$N = N_0\, e^{-\lambda t}$, $t_{1/2} = \ln 2/\lambda$
Photon energy	$E = hc/\lambda = c^6 / (2\pi G \nu_e H \lambda)$
Rydberg constant	$R_\infty = \pi^3 m_e e^4 G^3 \nu_e^3 H^3 / (\varepsilon_0^2 c^{16})$

Bond energy → frequency	$\nu = \Delta E/h = 2\pi\,\Delta E\,G\,\nu_e\,H / c^5$

4. Elements as ether structures 1–118

Structure number = atomic number. Frequency $f = mc^2/h$. Ladder position $k = \log(m/m_U)/\log(m_P/m_U)$, with $m_U = hH/c^2$ and m_P the Planck mass. Ether form: $f = H^{1-k/2} \cdot \nu_e^{k/2}$ (exact). Values for $H_0 = 62.0$.

Z	Symbol	Atomic mass (u)	f (Hz)	k	Ether form
1	H	1.008	2.270×10^{23}	0.6823	$H^{0.6588}.\nu_e^{0.3412}$
2	He	4.0026	9.015×10^{23}	0.6923	$H^{0.6539}.\nu_e^{0.3461}$
3	Li	6.94	1.563×10^{24}	0.6962	$H^{0.6519}.\nu_e^{0.3481}$
4	Be	9.0122	2.030×10^{24}	0.6981	$H^{0.6509}.\nu_e^{0.3491}$
5	B	10.81	2.435×10^{24}	0.6994	$H^{0.6503}.\nu_e^{0.3497}$
6	C	12.011	2.705×10^{24}	0.7002	$H^{0.6499}.\nu_e^{0.3501}$
7	N	14.007	3.155×10^{24}	0.7013	$H^{0.6493}.\nu_e^{0.3507}$
8	O	15.999	3.604×10^{24}	0.7023	$H^{0.6489}.\nu_e^{0.3511}$
9	F	18.998	4.279×10^{24}	0.7035	$H^{0.6482}.\nu_e^{0.3518}$
10	Ne	20.18	4.545×10^{24}	0.7039	$H^{0.6480}.\nu_e^{0.3520}$
11	Na	22.99	5.178×10^{24}	0.7049	$H^{0.6476}.\nu_e^{0.3524}$
12	Mg	24.305	5.474×10^{24}	0.7053	$H^{0.6474}.\nu_e^{0.3526}$
13	Al	26.982	6.077×10^{24}	0.7060	$H^{0.6470}.\nu_e^{0.3530}$
14	Si	28.085	6.326×10^{24}	0.7063	$H^{0.6468}.\nu_e^{0.3532}$
15	P	30.974	6.976×10^{24}	0.7070	$H^{0.6465}.\nu_e^{0.3535}$
16	S	32.06	7.221×10^{24}	0.7073	$H^{0.6464}.\nu_e^{0.3536}$
17	Cl	35.45	7.985×10^{24}	0.7080	$H^{0.6460}.\nu_e^{0.3540}$
18	Ar	39.95	8.998×10^{24}	0.7089	$H^{0.6456}.\nu_e^{0.3544}$
19	K	39.098	8.806×10^{24}	0.7087	$H^{0.6456}.\nu_e^{0.3544}$
20	Ca	40.078	9.027×10^{24}	0.7089	$H^{0.6455}.\nu_e^{0.3545}$
21	Sc	44.956	1.013×10^{25}	0.7097	$H^{0.6451}.\nu_e^{0.3549}$
22	Ti	47.867	1.078×10^{25}	0.7102	$H^{0.6449}.\nu_e^{0.3551}$
23	V	50.942	1.147×10^{25}	0.7106	$H^{0.6447}.\nu_e^{0.3553}$
24	Cr	51.996	1.171×10^{25}	0.7108	$H^{0.6446}.\nu_e^{0.3554}$
25	Mn	54.938	1.237×10^{25}	0.7112	$H^{0.6444}.\nu_e^{0.3556}$

26	Fe	55.845	1.258×10^{25}	0.7113	$H^{0.6444}.v_e^{0.3556}$
27	Co	58.933	1.327×10^{25}	0.7117	$H^{0.6442}.v_e^{0.3558}$
28	Ni	58.693	1.322×10^{25}	0.7117	$H^{0.6442}.v_e^{0.3558}$
29	Cu	63.546	1.431×10^{25}	0.7122	$H^{0.6439}.v_e^{0.3561}$
30	Zn	65.38	1.473×10^{25}	0.7124	$H^{0.6438}.v_e^{0.3562}$
31	Ga	69.723	1.570×10^{25}	0.7129	$H^{0.6436}.v_e^{0.3564}$
32	Ge	72.63	1.636×10^{25}	0.7132	$H^{0.6434}.v_e^{0.3566}$
33	As	74.922	1.688×10^{25}	0.7134	$H^{0.6433}.v_e^{0.3567}$
34	Se	78.971	1.779×10^{25}	0.7138	$H^{0.6431}.v_e^{0.3569}$
35	Br	79.904	1.800×10^{25}	0.7139	$H^{0.6431}.v_e^{0.3569}$
36	Kr	83.798	1.887×10^{25}	0.7142	$H^{0.6429}.v_e^{0.3571}$
37	Rb	85.468	1.925×10^{25}	0.7144	$H^{0.6428}.v_e^{0.3572}$
38	Sr	87.62	1.974×10^{25}	0.7145	$H^{0.6427}.v_e^{0.3573}$
39	Y	88.906	2.002×10^{25}	0.7147	$H^{0.6427}.v_e^{0.3573}$
40	Zr	91.224	2.055×10^{25}	0.7148	$H^{0.6426}.v_e^{0.3574}$
41	Nb	92.906	2.093×10^{25}	0.7150	$H^{0.6425}.v_e^{0.3575}$
42	Mo	95.95	2.161×10^{25}	0.7152	$H^{0.6424}.v_e^{0.3576}$
43	Tc	98	2.207×10^{25}	0.7154	$H^{0.6423}.v_e^{0.3577}$
44	Ru	101.07	2.276×10^{25}	0.7156	$H^{0.6422}.v_e^{0.3578}$
45	Rh	102.91	2.318×10^{25}	0.7157	$H^{0.6421}.v_e^{0.3579}$
46	Pd	106.42	2.397×10^{25}	0.7159	$H^{0.6420}.v_e^{0.3580}$
47	Ag	107.87	2.430×10^{25}	0.7160	$H^{0.6420}.v_e^{0.3580}$
48	Cd	112.41	2.532×10^{25}	0.7163	$H^{0.6418}.v_e^{0.3582}$
49	In	114.82	2.586×10^{25}	0.7165	$H^{0.6418}.v_e^{0.3582}$
50	Sn	118.71	2.674×10^{25}	0.7167	$H^{0.6416}.v_e^{0.3584}$
51	Sb	121.76	2.742×10^{25}	0.7169	$H^{0.6415}.v_e^{0.3585}$
52	Te	127.6	2.874×10^{25}	0.7173	$H^{0.6414}.v_e^{0.3586}$
53	I	126.9	2.858×10^{25}	0.7172	$H^{0.6414}.v_e^{0.3586}$
54	Xe	131.29	2.957×10^{25}	0.7175	$H^{0.6413}.v_e^{0.3587}$
55	Cs	132.91	2.994×10^{25}	0.7176	$H^{0.6412}.v_e^{0.3588}$

56	Ba	137.33	3.093×10^{25}	0.7178	$H^{0.6411}.v_e^{0.3589}$
57	La	138.91	3.129×10^{25}	0.7179	$H^{0.6411}.v_e^{0.3589}$
58	Ce	140.12	3.156×10^{25}	0.7179	$H^{0.6410}.v_e^{0.3590}$
59	Pr	140.91	3.174×10^{25}	0.7180	$H^{0.6410}.v_e^{0.3590}$
60	Nd	144.24	3.249×10^{25}	0.7181	$H^{0.6409}.v_e^{0.3591}$
61	Pm	145	3.266×10^{25}	0.7182	$H^{0.6409}.v_e^{0.3591}$
62	Sm	150.36	3.387×10^{25}	0.7184	$H^{0.6408}.v_e^{0.3592}$
63	Eu	151.96	3.423×10^{25}	0.7185	$H^{0.6407}.v_e^{0.3593}$
64	Gd	157.25	3.542×10^{25}	0.7188	$H^{0.6406}.v_e^{0.3594}$
65	Tb	158.93	3.580×10^{25}	0.7188	$H^{0.6406}.v_e^{0.3594}$
66	Dy	162.5	3.660×10^{25}	0.7190	$H^{0.6405}.v_e^{0.3595}$
67	Ho	164.93	3.715×10^{25}	0.7191	$H^{0.6404}.v_e^{0.3596}$
68	Er	167.26	3.767×10^{25}	0.7192	$H^{0.6404}.v_e^{0.3596}$
69	Tm	168.93	3.805×10^{25}	0.7193	$H^{0.6404}.v_e^{0.3596}$
70	Yb	173.05	3.898×10^{25}	0.7195	$H^{0.6403}.v_e^{0.3597}$
71	Lu	174.97	3.941×10^{25}	0.7195	$H^{0.6402}.v_e^{0.3598}$
72	Hf	178.49	4.020×10^{25}	0.7197	$H^{0.6402}.v_e^{0.3598}$
73	Ta	180.95	4.076×10^{25}	0.7198	$H^{0.6401}.v_e^{0.3599}$
74	W	183.84	4.141×10^{25}	0.7199	$H^{0.6401}.v_e^{0.3599}$
75	Re	186.21	4.194×10^{25}	0.7200	$H^{0.6400}.v_e^{0.3600}$
76	Os	190.23	4.285×10^{25}	0.7201	$H^{0.6399}.v_e^{0.3601}$
77	Ir	192.22	4.329×10^{25}	0.7202	$H^{0.6399}.v_e^{0.3601}$
78	Pt	195.08	4.394×10^{25}	0.7203	$H^{0.6398}.v_e^{0.3602}$
79	Au	196.97	4.436×10^{25}	0.7204	$H^{0.6398}.v_e^{0.3602}$
80	Hg	200.59	4.518×10^{25}	0.7205	$H^{0.6397}.v_e^{0.3603}$
81	Tl	204.38	4.603×10^{25}	0.7207	$H^{0.6397}.v_e^{0.3603}$
82	Pb	207.2	4.667×10^{25}	0.7208	$H^{0.6396}.v_e^{0.3604}$
83	Bi	208.98	4.707×10^{25}	0.7208	$H^{0.6396}.v_e^{0.3604}$
84	Po	209	4.707×10^{25}	0.7208	$H^{0.6396}.v_e^{0.3604}$
85	At	210	4.730×10^{25}	0.7209	$H^{0.6396}.v_e^{0.3604}$

86	Rn	222	5.000×10^{25}	0.7213	$H^{0.6394}.v_e^{0.3606}$
87	Fr	223	5.023×10^{25}	0.7213	$H^{0.6394}.v_e^{0.3606}$
88	Ra	226	5.090×10^{25}	0.7214	$H^{0.6393}.v_e^{0.3607}$
89	Ac	227	5.113×10^{25}	0.7214	$H^{0.6393}.v_e^{0.3607}$
90	Th	232.04	5.226×10^{25}	0.7216	$H^{0.6392}.v_e^{0.3608}$
91	Pa	231.04	5.204×10^{25}	0.7215	$H^{0.6392}.v_e^{0.3608}$
92	U	238.03	5.361×10^{25}	0.7218	$H^{0.6391}.v_e^{0.3609}$
93	Np	237	5.338×10^{25}	0.7217	$H^{0.6391}.v_e^{0.3609}$
94	Pu	244	5.496×10^{25}	0.7219	$H^{0.6390}.v_e^{0.3610}$
95	Am	243	5.473×10^{25}	0.7219	$H^{0.6390}.v_e^{0.3610}$
96	Cm	247	5.563×10^{25}	0.7220	$H^{0.6390}.v_e^{0.3610}$
97	Bk	247	5.563×10^{25}	0.7220	$H^{0.6390}.v_e^{0.3610}$
98	Cf	251	5.653×10^{25}	0.7221	$H^{0.6389}.v_e^{0.3611}$
99	Es	252	5.676×10^{25}	0.7222	$H^{0.6389}.v_e^{0.3611}$
100	Fm	257	5.789×10^{25}	0.7223	$H^{0.6388}.v_e^{0.3612}$
101	Md	258	5.811×10^{25}	0.7223	$H^{0.6388}.v_e^{0.3612}$
102	No	259	5.834×10^{25}	0.7224	$H^{0.6388}.v_e^{0.3612}$
103	Lr	266	5.991×10^{25}	0.7226	$H^{0.6387}.v_e^{0.3613}$
104	Rf	267	6.014×10^{25}	0.7226	$H^{0.6387}.v_e^{0.3613}$
105	Db	268	6.036×10^{25}	0.7226	$H^{0.6387}.v_e^{0.3613}$
106	Sg	269	6.059×10^{25}	0.7226	$H^{0.6387}.v_e^{0.3613}$
107	Bh	270	6.081×10^{25}	0.7227	$H^{0.6387}.v_e^{0.3613}$
108	Hs	269	6.059×10^{25}	0.7226	$H^{0.6387}.v_e^{0.3613}$
109	Mt	278	6.262×10^{25}	0.7229	$H^{0.6386}.v_e^{0.3614}$
110	Ds	281	6.329×10^{25}	0.7230	$H^{0.6385}.v_e^{0.3615}$
111	Rg	282	6.352×10^{25}	0.7230	$H^{0.6385}.v_e^{0.3615}$
112	Cn	285	6.419×10^{25}	0.7231	$H^{0.6385}.v_e^{0.3615}$
113	Nh	286	6.442×10^{25}	0.7231	$H^{0.6385}.v_e^{0.3615}$
114	Fl	289	6.509×10^{25}	0.7232	$H^{0.6384}.v_e^{0.3616}$
115	Mc	290	6.532×10^{25}	0.7232	$H^{0.6384}.v_e^{0.3616}$

116	Lv	293	6.599×10^{25}	0.7233	$H^{0.6384}\cdot v_e^{0.3616}$
117	Ts	294	6.622×10^{25}	0.7233	$H^{0.6384}\cdot v_e^{0.3616}$
118	Og	294	6.622×10^{25}	0.7233	$H^{0.6384}\cdot v_e^{0.3616}$

5. Status

Predictions P1–P5 and the disclosure of data already examined are stated in the preregistration text. This compendium adds no new prediction; it records the formulas and present values as of 2026-11-05.